

EXPERIENCE

Google

US-MTV

• *Software Engineer, Site Reliability Engineering*

Mar '19 - Present

- **Gemini Platform & Vertex AI:** Making Google Cloud's Large Language Model offerings (Gemini, Veo, Imagen, etc) more reliable. Check out cloud.google.com/vertex-ai
 - * Reliability owner for **accelerator-backed inference serving** at frontier-model scale — GPU/TPU fleet capacity, serving-stack rollouts, and the incident path for Gemini, Veo and Imagen traffic.
 - * Led the migration of Vertex AI's **ML model-serving configurations** to dedicated hermetic ML-Ops, removing a class of release-time serving outages and materially improving rollout safety.
 - * Automated the months-long turnup of **Vertex AI** in new regions, including accelerator capacity validation before traffic admission.
 - * Built a **Gemini-powered incident framework** that cuts MTTx through LLM-assisted triage and gated actuation — an agentic system running against live production.
- **Serverless Platform:** Making Google Cloud's Serverless compute offerings (App Engine, Cloud Functions, Cloud Run) more reliable. Check out cloud.google.com/serverless
 - * Automated months-long process of turning up **Cloud Run** in new regions.
 - * Enabled safe, progressive rollouts of schema and config changes across **560+ Spanner databases**, eliminating the prior failure mode where bad changes affected customers globally for multiple hours.
 - * Built an *actionable-metrics framework* that democratized signal across SRE and partner teams, driving a **50%** reduction in oncall load while the team onboarded **40%** more partner services.
 - * Drove down resource ceilings across all Serverless Bigtables by adopting *Autocap*, materially improving fleet-wide compute efficiency.
 - * Led the **Log4J** code-red response across four Serverless products under incident pressure.
 - * Led **4 interns** to build *Khoj*, a Google-wide automated incident root-causing system that **reduced mean time to response and mitigation from multiple hours to just minutes**.
- **Monitoring & Alerting:** Making Google Cloud's monitoring and alerting more reliable.
 - * Seamlessly migrated fragmented internal users from 15-year-old + 4-year-old alert visualization tools to the next-gen tool by adopting the Google-wide experiment framework.
 - * Shipped the *Alert2Bug* migration in **1 week** against a multi-month estimate by leaning on existing primitives instead of a bespoke rebuild.

Booking.com

NL-AMS

• *Software Developer*

Jun '17 - Feb '19

- **Image Services:** Built a Java web service to serve images on-the-fly throughout **booking.com**. Moved away from the legacy pipeline to reduce storage consumption while implementing an ORM layer for the service to easily fetch images from various data sources. Successfully reduced image generation latency by over **50%**.

AGENT HARNESSES & EVALUATION INFRASTRUCTURE (INDEPENDENT)

- **Multi-agent fleet harness:** Orchestration layer running a persistent fleet of coding agents across two machines: **40** operator entrypoints, **23** hermetic test suites, and a typed Python core held under **mypy --strict**. Built on a **generator ≠ evaluator** rail — an independent evaluator agent judges a change by *executing* it (runs the suite, mutates the code) rather than by reading the diff, and no change merges past a live REJECT. Uses **mutation testing as the primary defect-finder**: a surviving mutant is a finding, not a curiosity.
- **Autonomous trading system:** Human-in-the-loop trading and portfolio engine across 10 broker integrations — **540** source files, **~188k lines of Python**, **15,001 tests at 82% coverage**, **58** MCP tools. Every capital-moving path sits behind an explicit approval gate; reliability work includes circuit breakers, a dead-letter queue, SLOs, and drawdown kill-switches. Deployed on Cloud Run via a health-gated build pipeline.
- **Fleet operations console:** Next.js/TypeScript console for live agent telemetry, approvals and cost attribution — **632** modules against **208** test files, with a typed API contract whose drift from the generated client fails CI.

ACHIEVEMENTS

- **ICPC 2016:** Team *YeToBadaToingHai* - **9th** out of over 3000 teams in Kharagpur Online; **32nd** out of over 3000 teams in Amritapuri Online.
- **CodeChef:** A **top 20** and numerous top **150** finishes in the monthly Long Challenge.
- **National Talent Search Exam 2009:** Scholarship awarded to top **0.1%** among 1 million students for excellence in scholastic and mental aptitude.

EDUCATION

- **International Institute of Information Technology** IN-HYD
Bachelor of Technology in Computer Science and Engineering *Aug 2013 – July 2017*
 - **All India Engineering Entrance Exam, 2013:** Secured **AIR 719**, Top **0.05%** among 1.4 million students

SKILLS

- **Languages:** Daily: Python (typed, `mypy --strict`), Go, bash, SQL, TypeScript; Prior production: C++, Java; Basic: Scala, Rust
- **ML & agent infrastructure:** GPU/TPU fleet operations, model-serving stacks, LLM evaluation harnesses, mutation testing, MCP tool servers, multi-agent orchestration
- **Platform:** Kubernetes, Terraform, Bazel, Prometheus, Spanner, BigQuery, ClickHouse, Cloud Run, GCS, launchd, git